INCH-POUND

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27 January 1997
SUPERSEDING
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1 October 1982
MS22012D
20 November 1969

PERFORMANCE SPECIFICATION

VALVE, HIGH-PRESSURE OXYGEN CYLINDER, AUTOMATIC OPENING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements of an automatic opening cylinder valve for use in aircraft high-pressure oxygen systems.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Air Warfare Center Aircraft Division, Code 414100B120-3, Highway 547, Lakehurst, NJ 08733-5100, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4820

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2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2d).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-O-27210	-	Oxygen, Aviator's Breathing, Liquid and Gas
MIL-G-27617	-	Grease, Aircraft and Instrument, Fuel and Oxidizer Resistant
MIL-T-27730	-	Tape, Antiseize, Tetrafluoroethylene with Dispenser

STANDARDS

FEDERAL

FED-STD-H28	-	Screw Thread Standards For Federal Services

DEPARTMENT OF DEFENSE

MIL-STD-130	-	Identification Marking of U. S. Military Property
MIL-STD-889	-	Dissimilar Metals
MIL-STD-1359	-	Cleaning Methods and Procedures For Breathing Oxygen Equipment
MS24392	-	Nipple, Tube, Precision Type
MS16624	-	Ring, Retaining, External, Basic (Taper Section Type)

AIR FORCE - NAVY AERONAUTICAL

AN818	-	Nut Tube Coupling, Short

(Unless otherwise indicated, copies of the above specifications and standards are available from the Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME - B46.1	-	Surface Texture (Surface Roughness, Waviness, and Lay) (DoD adopted)
ASME - Y14.5	-	Dimensioning and Tolerancing

(Application for copies should be addressed to the American Society Of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN NATIONAL STANDARD INSTITUTE (ANSI)

ANSI- Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes (DoD adopted)

(Application for copies should be addressed to American National Standard Institute, 11 West, 42nd Street, New York, NY 10036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D1149	-	Rubber Deterioration, Surface Ozone Cracking in a Chamber Standard Test Method for (DoD adopted)
ASTM-G63	-	Nonmetallic Materials for Oxygen Service Standard Guide for Evaluating
ASTM-G72	-	Standard Test Method For Autogeneous Ignition Temperature of Liquids and Solids in a High Pressure Oxygen Enriched Environment
ASTM-G94	-	Metals for Oxygen Service, Standard Guide for Evaluating

(Application for copies should be addressed to the American Society for Testing and Materials 100 Barr Harbor Drive, Philadelphia, PA 19103.)

2.4 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. Materials used shall be selected from the applicable government or non-

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Government specifications and standards. Materials selected shall be compatible with oxygen fluid and shall meet the requirements of this specification.

3.2.1 Metal parts. All metal parts shall be of corrosion-resistant material or treated to resist to corrosion. Material for metal parts shall be selected in accordance with ASTM-G94 for oxygen compatibility of metals.

3.2.1.1 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used. The dissimilar metals shall be in accordance with MIL-STD-889.

3.2.2 Nonmetallic materials. Nonmetallic materials that are adversely affected by continued use with oxygen shall not be used. Material for nonmetallic parts shall be selected in accordance with ASTM-G63 for oxygen compatibility of nonmetallic materials. Nonmetallic parts, wetted by oxygen, shall be resistant to auto-ignition, and shall have the lowest practicable heat of combustion in accordance with ASTM-G63. Additionally, where nonmetallic parts are installed, the design shall minimize the surface area and volume exposed to oxygen.

3.2.2.1 Age control. Because of the mechanical properties of elastomer are sensitive to temperatures, elastomer components except silicones shall be treated for protection from the environmental conditions such as moisture and extreme temperatures. The integral elastomer components shall be operable throughout the life of the valve. The shelf life of the elastomer components shall be determined by the acquisition activities (see 6.2).

3.2.2.2 Elastomer components. Elastomer components shall be composed of an ozone-resistant composition which shall not swell and shall meet the ozone-resistant requirements specified herein (see 4.5.12).

3.2.3 Protective treatment of materials. Materials used shall be protected against environmental deterioration. Protective coatings that may crack, chip, or scale during normal service life or under extreme environmental conditions shall not be used.

3.2.4 Surface finish. All machined surfaces shall be smooth, free of pits and discontinuities; such as, scratches or breaks and shall be uniform in appearance. Surface roughness shall be in accordance with ASME-B46.1.

3.2.5 Lubricants. Lubricants shall not be used within the normal flow path of oxygen. Lubricants shall be in accordance with MIL-G-27617, Type III and shall have been tested for compatibility with gaseous oxygen in accordance with ASTM-G72.

3.3 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials shall be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Design and interface requirements. The design and interface requirements of the valve shall

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be as specified in 3.4.1 through 3.4.10. The construction of the valve shall be such that it opens automatically when the coupling nut is connected to the valve outlet and closes automatically when disconnected.

3.4.1 Outlet. The valve outlet port shall be provided with a protective device for protecting threads and preventing entrance of contaminants.

3.4.2 Siphon tube. The valve shall be provided with a siphon tube or similar device to prevent any contaminants within the cylinder from being carried into the valve.

3.4.3 Safety features. Valves shall be provided with a safety outlet which shall connect directly with a valve inlet. The safety outlet shall include means for rapid pressure dumping.

3.4.4 Valve assembly. The valve assembly shall consist of a valve and a coupling assembly. The dimensions, physical characteristics, and interface requirements of the valve assembly and integral components shall be in accordance with table I.

Table I. Dimensions and configurations of the valve assembly and integral components.

1	Valve assembly	Figure 1
2	Nose adapter	Figure 2
3	Coupling nut	Figure 3
4	Cylinder valve	Figure 4

3.4.5 Coupling assembly. The coupling assembly shall consist of a nose adapter, a coupling nut, a packing ring, and an MS16624 retaining ring.

3.4.6 Check valve. The valve shall incorporate a check valve or similar device to prevent oxygen leakage when the nose adapter is uncoupled from the valve.

3.4.7 Antiseize tape. Polytetrafluorotheylene tape shall be used on all pipe thread fittings. The tape shall be applied in accordance with the applicable instruction of MIL-T-27730 to minimize potential contamination of the breathing oxygen systems and provides an adequate seal. Tape shall not be used on flare tube fitting, straight threads coupling sleeves, or outside of tube flares.

3.4.8 Cleanliness. Prior to assembling the valve and its subassemblies, all internal and external surfaces shall be cleaned in accordance with MIL-STD-1359 to ensure that the valve is free of any contamination.

3.4.9 Threads. To be interfaced with the valve assembly installed in the aircraft oxygen systems, screw threads and pipe threads used in construction of valves shall be in accordance with

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applicable FED-STD-H28 (see figures 1, 3, and 4).

3.4.10 Dimensioning and tolerancing. Dimensioning and tolerancing of the valve shall be in accordance with ASME - Y14.5.

3.5 Performance.

3.5.1 Leakage. The valve shall not leak more than 2 milliliters (ml) per minute when tested as specified in 4.5.2.

3.5.2 Pressure drop. The pressure drop across the valve shall be not greater than 5 psig when tested as specified in 4.5.3.

3.5.3 Proof pressure. The proof pressure of the valve shall not leak more than 2 ml per minute when tested as specified in 4.5.4.

3.5.4 High temperature operation. The valve shall not show any evidence of mechanical or material failure after being stabilized and tested as specified in 4.5.5.

3.5.5 Low temperature operation. The valve shall not show any evidence of mechanical or material failure after being stabilized and tested as specified in 4.5.6.

3.5.6 Vibration. The valve shall not show any evidence of mechanical or material failure after tested as specified in 4.5.7.

3.5.7 Endurance. The valve shall not show any evidence of mechanical or material failure after tested as specified in 4.5.8.

3.5.8 Torque.

3.5.8.1 Opening. The valve shall open fully when tested as specified in 4.5.9.1. The flow rate of valve at fully open position is 50-65 liters per minute.

3.5.8.2 Sealing. The valve shall not leak in excess of 2 ml per minute when tested as specified in 4.5.9.2.

3.5.9 Check valve. The check valve shall not leak in excess of 2 ml per minute when tested as specified in 4.5.10.

3.5.10 Safety disk rupture. The safety disk shall rupture between 2,700 and 3,000 psig when tested as specified in 4.5.11.

3.5.11 Ozone resistance. The elastomer components used shall be resistant to ozone. The elastomer components shall not show any evidence of cracking or damage when tested as specified in 4.5.12.

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3.5.12 Oxygen test.

3.5.12.1 Oxygen ignition. The nonmetallic material and valve shall not ignite and there shall be no evidence of charring or deterioration when tested as specified in 4.5.13.

3.5.12.2 Oxygen purge. The valve shall not ignite and there shall be not any evidence of charring or deterioration when tested as specified in 4.5.13.2.

3.5.13 Disassembly and examination. The valve shall show no evidence of deterioration or wear due to testing that will affect the valve performance when examined as specified in 4.5.14.

3.6 Interchangeability. All parts having the same manufacturer's part number shall be interchangeable functionally and dimensionally.

3.7 Weight. The total weight of the valve assembly, exclusive of the outlet cap, shall be not greater than 14 ounces.

3.8 Identification of product. The valves shall be marked for identification in accordance with MIL-STD-130 except that the national stock number shall be omitted.

3.9 Superseding part numbers. This valve assembly supersedes the MS22012 valve assembly. The new part numbers which supersedes the corresponding old MS part numbers are as shown in table II.

3.10 Workmanship. The valves shall be uniform in quality and shall be free from irregularities, defects or foreign material which could adversely affect safety, performance, reliability, or durability.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follow**:**

a. First article inspection (see 4.2)

b. Conformance inspection (see 4.3)

4.2 First article inspection. First article inspection consists of examination and test of the production item after award of a contract to determine the production item conforms to the requirements of this specification. First article inspection shall consist of all examinations and tests of this specification.

4.2.1 First article samples. Unless otherwise specified (see 6.2), the manufacturer shall submit 4 valves and 6 elastomer test slabs, 5 inches by 6 inches by 0.075 inch, composed of identical ozone-resistant composition and cure that was used in manufacturing the elastomer components.

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The samples shall be representative of the construction, workmanship, components, and materials to be used during production. Approval of the first article inspection samples or the waiving of the first article inspection does not exempt the test requirements for performing the conformance inspection. The first article inspection samples shall be furnished to the Government as directed by the acquisition activity (see 6.2).

Table II. Superseding part number.

Items	New Part Number	Canceled Part Number
Valve assembly	M8622-1	MS22012-1
Cylinder valve	M8622-2	MS22012-2
Coupling assembly	M8622-3	MS22012-3
Nose adapter	M8622-4	MS22012-4
Coupling nut	M8622-5	MS22012-5
Packing ring	M8622-6	MS22012-6

4.3. Conformance inspection. Conformance inspection shall consist of examinations and tests as specified in table III. The tests shall perform on individual products or lots to determine conformance of the products or lots with the requirements of this specification. The sampling and inspection shall conform to ANSI-Z1.4.

4.3.1 Sampling.

4.3.1.1 Inspection lot.

4.3.1.1.1 Valve. An inspection lot size shall be expressed in units of valves made under the same conditions and from the same materials and components. The sample unit shall be one valve.

4.3.1.2 Sampling for tests and examinations of valves. The sample size, acceptance criteria, tests and examinations required for the valves shall be as specified in table III.

4.4 Test conditions.

4.4.1 Oxygen. Unless otherwise specified (see 6.2), the oxygen used in testing the valves shall conform to MIL-O-27210, Type I. If oxygen is not used as the test gas, oxygen in accordance with MIL-O-27210, Type I shall be used to purge in accordance with 3.5.12.2 and 4.5.13.2.

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TABLE III. Sample size, acceptance criteria, examinations, and tests of the valve.

Inspection	Method	Sample size	Acceptance criteria
Visual Examination (see Table IV for classification of defects)	4.5.1.1	Every valve for critical defects Inspection level for minor defects to be determined by the acquisition activities	An acceptable quality level to be determined by the acquisition activity
Dimensions	4.5.1.1.1	Inspection level to be determined by the acquisition activity	Acceptable quality level to be determined by the acquisition activity
Leakage	4.5.2	Every valve	Acceptable quality level to be determined by the acquisition activity
Pressure drop	4.5.3	Every valve	Acceptable quality level to be determined by the acquisition activity
Proof pressure	4.5.4	Every valve	Acceptable quality level to be determined by the acquisition activity
Opening	4.5.9.1	Every valve	Acceptable quality level to be determined by the acquisition activity
Identification of product	4.5.1.1.2	Every valve	Acceptable quality level to be determined by the acquisition activity
Oxygen purge	4.5.13.2	Every valve	Acceptable quality level to be determined by the acquisition activity

4.4.2 Temperature and pressure. Unless otherwise specified (see 6.2), tests shall be conducted at local ambient temperature and barometric pressure. The temperature and barometric pressure shall be recorded at the time of inspection. This information shall be available for computation of test data, where required, at normal temperature and pressure (NTP) condition. NTP conditions are 70 ^{0}F (21.1 ^{0}C) and 29.2 inches of mercury (101.3 kPa). Test instruments shall be calibrated or adjusted according to their required usage in conducting individual tests.

4.5 Inspection methods.

4.5.1 Visual examination.

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4.5. l.1 Valves. Every valve shall be examined visually for critical defects. Every valve selected as a sample unit from the lot shall be visually examined for defects to determine conformance to this specification. The classification of defects is specified in table IV.

4.5.1.1.1 Dimensions. Every valve selected as a sample unit from the lot shall be checked dimensionally to determine conformance to the dimensions specified in figure 1 through figure 4.

4.5.1.1.2 Identification of product. Every valve selected from a sample lot shall be examined for defects listed in table III to determine the identification of product or marking shall conform to 3.8.

4.5.2 Leakage. With the nose adapter of the valve uncoupled, a gauge pressure of 15 psig shall be applied to the valve inlet for a period of 2 minutes. This procedure shall than be repeated utilizing a pressure of 2,000 psig. During these periods the valve shall meet the requirements specified in 3.5.1. The nose adapter shall then be coupled to the valve and the adapter outlet plugged. A pressure of 2,000 psig shall be applied to the coupled assembly for a period of 2 minutes. During this period, the coupled assembly shall meet the requirements specified in 3.5.1.

4.5.3 Pressure drop. A constant pressure shall be applied to the inlet and a flow of 55 liters per minute maintained at the outlet. Under these conditions, the pressure drop across the valve shall be measured. The valve shall pass the requirements specified in 3.5.2.

4.5.4 Proof pressure. The nose adapter shall be uncoupled from the valve. Removed the valve outlet cap and the frangible disk if used, and the outlet port shall be replaced with a solid plug. A hydraulic pressure of 3,000 psig shall then be applied to the inlet of the valve for 5 minutes. During this period, the valve shall pass the requirements specified in 3.5.3.

4.5.5 High temperature operation. The valve shall be installed in the test chamber and the temperature shall be raised to 160 $\pm$ 5 ^{o}F (71.1 $\pm$ 2.8 ^{o}C) and maintained for 3 hours. After the 3-hours stabilization period and while at this temperature, the valve shall be subjected to and pass the leakage and pressure drop tests (see 4.5.2 and 4.5.3). The valve shall pass the requirements specified in 3.5.4.

4.5.6 Low temperature operation. The valve shall be installed in a test chamber and the temperature shall be lowered down to -65 $\pm$ 5 ^{0}F (-53.9 $\pm$ 2.8 ^{0}C) and maintained for 3 hours. After the 3 hour stabilization period and while at this temperature, the valve shall be subjected to and pass the leakage and pressure drop tests. The valve shall pass the requirements specified in 3.5.5.

4.5.7 Vibration. The nose adapter shall be coupled to the valve, the adapter outlet shall be plugged, and the frangible disk shall be placed with a solid plug. A pressure of 2,000 psig shall be applied to the outlet end of the valve. The valve shall then be continuously vibrated for a period of 6 hours using a double amplitude of 0.018 to 0.023 inch at a varying frequency of 300 to 3,000 cpm. Every 2 hour during the vibration period and upon completion of the vibration period, the valve shall be subjected to and pass the leakage and pressure drop tests. The valve

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shall pass the requirements specified in 3.5.6.

TABLE IV. Classification of defects for visual examination of valves.

Critical	Minor
1. Material imperfections - foreign matter embedded	1 Marking missing, insufficient, incorrect, illegible, or not permanent.
2. Surface - unclean, rough, realigned, or contains cracks, nicks, or flaws	
3. Any component missing, malformed, fractured, or damaged	
4. Any component loose or not securely restrained	
5. Incorrect assembling or improper positioning of components	
6. Any functioning part that works with difficulty	
7. Faulty workmanship or other irregularities	

4.5.8 Endurance. The valve shall be connected to a 2,000 psig source of pressure. The coupling nut of the valve shall then be connected and disconnected to and from the valve. This shall constitute one cycle. During the cycling, the initial valve inlet pressure shall be 1,800 psig that shall be permitted to decay to a minimum of 300 psig. Flow rate through the valve during cycling shall be 50 liters per minute for 10 seconds minimum. There shall be a minimum of 5 seconds between each cycle. The valve shall be subjected to 250 cycles and shall pass the requirements specified in 3.5.7. At the conclusion of the test, the valve shall be subjected to and pass the leakage and pressure drop tests.

4.5.9 Torque.

4.5.9.1 Opening. A pressure of 1,800 psig shall be applied to the valve inlet. While maintaining this pressure, a torque of 35 inch-pounds maximum shall be applied to the valve coupling nut in the opening direction. During these conditions, the valve shall pass the requirements specified in 3.5.8.1.

4.5.9.2 Sealing. A pressure of 1,800 psig shall be applied to the valve inlet. While maintaining this pressure, a torque of 150 inch-pounds maximum shall be applied to the valve coupling nut. During these conditions, the valve shall pass the requirements specified in 3.5.8.2.

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4.5.10 Check valve. The nose adapter shall be uncoupled from the valve and an external pressure of 15 psig applied to the valve outlet. The valve shall pass the requirements specified in 3.5.9.

4.5.11 Safety disk rupture. The safety disk shall be installed in a test fixture and subjected to a uniformly applied hydrostatic pressure of 2,700 to 3,000 psig. The safety disk shall meet the requirements specified in 3.5.10. If the safety disk does not rupture at 3,000 psig for one minute, the test valve shall be rejected.

4.5.12 Ozone-resistance (elastomer components). Three of the six test slabs submitted for first article inspection shall be tested for ozone-resistance. The test apparatus shall be in accordance with ASTM-D1149. The test slabs shall be elongated 20 percent, placed in an ozone-free atmosphere for 24 hours, then placed in the ozone chamber. The chamber shall be adjusted to $100 \pm 2\ ^{0}F$ to give an exposure of ozone concentration of 120 ± 10 parts by volume of ozone per million parts by volume of air. The air-ozone velocity in the chamber shall be at least 2 feet per second. The material shall be exposed to these conditions for 60 minutes. The test slabs shall then be examined under 10X magnification and shall pass the requirements specified in 3.5.11.

4.5.13 Oxygen test.

4.5.13.1 Oxygen ignition. Nonmetallic parts and the valve shall be subjected to oxygen bomb tests to determine compatibility of the items with high pressure oxygen.

4.5.13.1.1 Test equipment. The test equipment used in performing these tests shall be in accordance with figures 5 and 6. The main valve shall be a quick-acting valve having a 0.25 inch opening, shall require no internal lubrication for operation, and shall be rated for at least 3,000 psig operation. The valve shall be leak tight when closed and shall open with such rapidity that, at least 95 percent of the total pressure rise shall have occurred within the test chamber in 20 ± 7 milliseconds. The pressure rise time shall be determined by measurement with a fast-responding small volume pressure transducer substituted for the test chamber in the test setup. The tubing and fitting connecting the valve to the test chamber shall be 0.188 inch size and shall be fabricated from corrosion-resistant steel. The tubing which run between the valve and the test chamber shall not be bent more than 4 degrees. The bleed valve shall use no organic material to accomplish its seal nor require interior lubrication for operation. The bleed valve shall be installed so that when it is closed, there is no flow of oxygen between the test chamber and the packing side of the valve. Test samples, 0.188 inch diameter by 0.075 inch thick shall be made from 3 of the test slabs submitted for first article inspection before the test slab is exposed to the ozone-resistance test.

4.5.13.1.2 Test procedure for elastomeric materials. A sample shall be inserted in the corrosion-resistant steel recess of the test chamber and the test chamber assembled. The quick-acting valve shall be fully opened and the test chamber pressurized with oxygen at $2,175 \pm 25$ psig for 30 seconds. At the end of the 30 second period the pressure in the test chamber shall be released by means of the bleed valve, the test chamber shall be opened, and the sample shall be visually examined. If there is no indication of ignition, pressurizing operation shall be repeated at 60 second intervals. The pressurizing operations shall be repeated until either ignition occurs or until three pressurizing operations have been performed. The sample shall be removed from the

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test chamber and examined for indication of charring or deterioration. The series of three pressurizing operations shall be repeated on a fresh sample. This procedure shall be continued until either a sample has ignited or until five test samples have been tested. The test material shall pass the requirements specified in 3.5.12.

4.5.13.1.3 Test procedure for valve. All connections on the valve except the high pressure inlet connection shall be sealed against a pressure of 2,000 psig. The high pressure inlet connection of the valve shall be installed in the test setup as shown on figure 1 in lieu of the test chamber. The quick-acting valve shall be fully opened and the check valve pressurized with oxygen to a pressure of 2,000 psig. This pressure shall be maintained for 30 seconds and then released through the bleed valve. The valve shall be disconnected and examined for indication of charring or deterioration of any nonmetallic material. The valve shall pass the requirements specified in 3.5.12.

4.5.13.2 Oxygen purge. The valve shall be connected to a regulated source of oxygen (4.4.1). An inlet pressure of 1,800 psig shall be applied and the valve shall free flow for 3 to 5 minutes. The valve shall pass the requirements specified in 3.5.12.2.

4.5.14 Disassembly and examination. At the conclusion of all the inspections, the valve shall be disassembled and examined. The valve shall be disassembled and examined after any test in which valve failure indicated faulty construction. The valve shall pass the requirements specified in 3.5.13.

4.5.15 Cleanliness. To ensure the valve is clean and free of all contamination, verification of cleanliness shall be tested in accordance with MIL-STD-1359.

5 PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's Systems Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The cylinder valves covered by this specification are used in filling and controlling the flow of oxygen from high pressure oxygen systems in aircraft.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Whether first article inspection is waived (see 4.2.1).
c. Part number or dash numbers required (see 3.9).
d. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1).
e. Name and address of the first article inspection laboratory (see 4.2.1).
f. Packaging requirements (see 5.1).
g. Marking requirements (see 4.5.1.1.2).
h. Age of elastomer component (see 3.2.2.1).

6.3. Statistical process control (SPC). Unless otherwise specified in the Statement Of Work (SOW), the contractor should implement and use statistical process control techniques in the manufacturing process for parts covered by this specification. SPC program should be controlled and developed and maintained in accordance with EIA-557 or equivalent. The SPC program should be documented and maintained as part of overall reliability assurance program. Evidence of such compliance should be verified by the acquisition activity of this specification.

ELECTRONIC INDUSTRY ASSOCIATION (EIA)

EIA-557 - EIA Standard Implementation of Statistical Process Control (SPC) in Manufacturing Process

(Application for copies should be forwarded to Electronic Industries Association, 2500 Wilson Boulevard, Arlington VA 22201-3834)

6.4 First article. If a first article inspection is required, specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article should be obtained from the contracting officer.

6.5 Subject term (key word) listing.

Aircraft system
Controlling gas
Crew system
Filling gas
Oxygen system

6.6 Change from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issues due to the extent of the changes.

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CONCLUDING MATERIAL

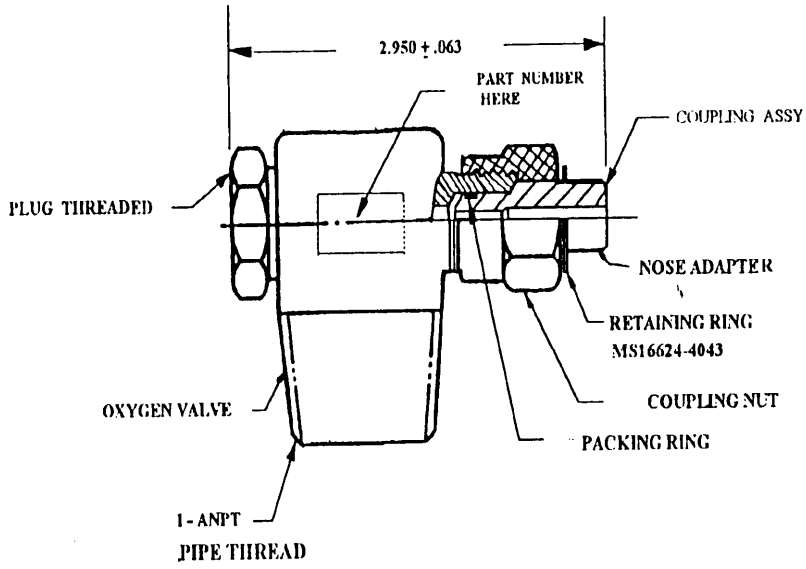
Custodians:
Army - AV
Navy -AS
Air Force - 99

Review activities:
Navy - SH, MC
Air Force - 11
DLA - CS

Preparing activity:
Navy -AS

(Project. 4820-0724)

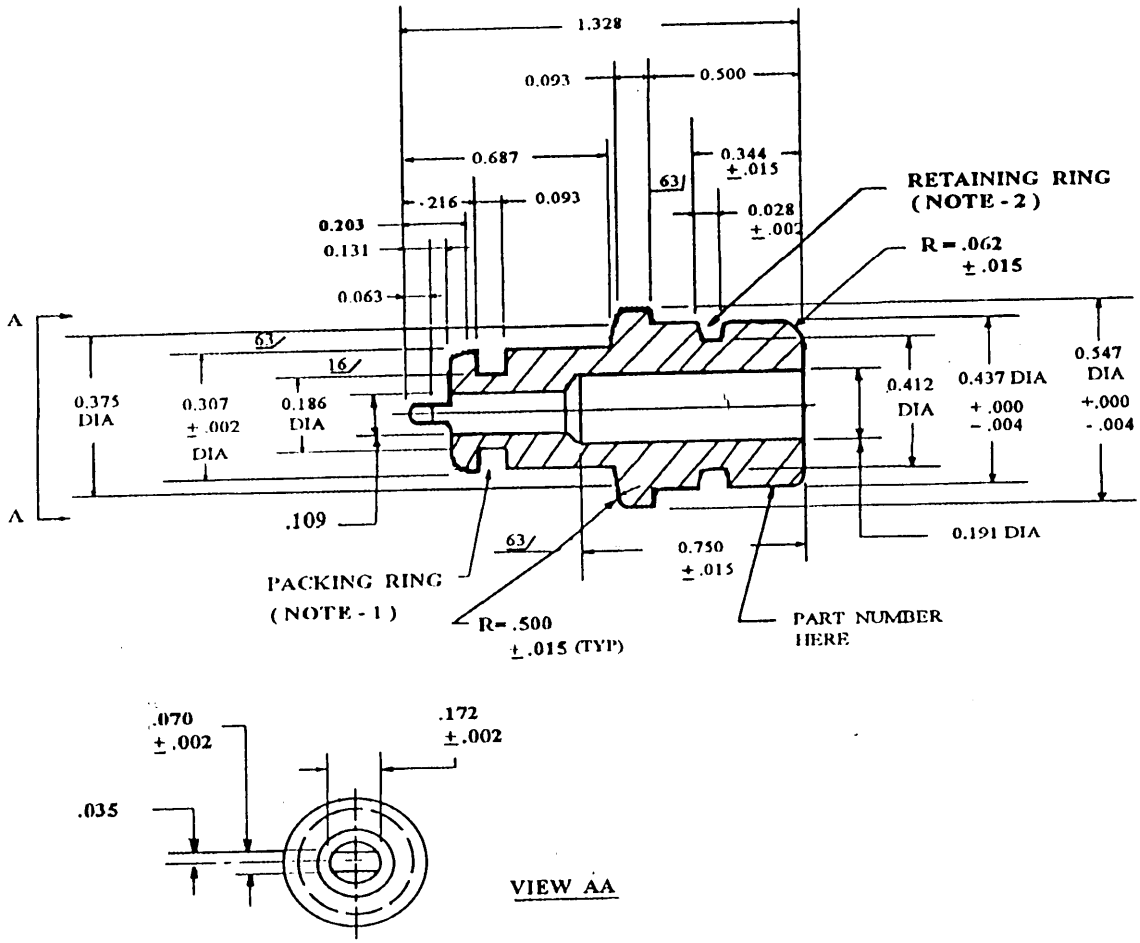
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Note: 1. AH dimensions are in inches,
2. Unless otherwise specified on the drawing, tolerances: decimal ± .00.5 and angle ± 1°.

FIGURE 1. Oxygen cylinder valve assembly.

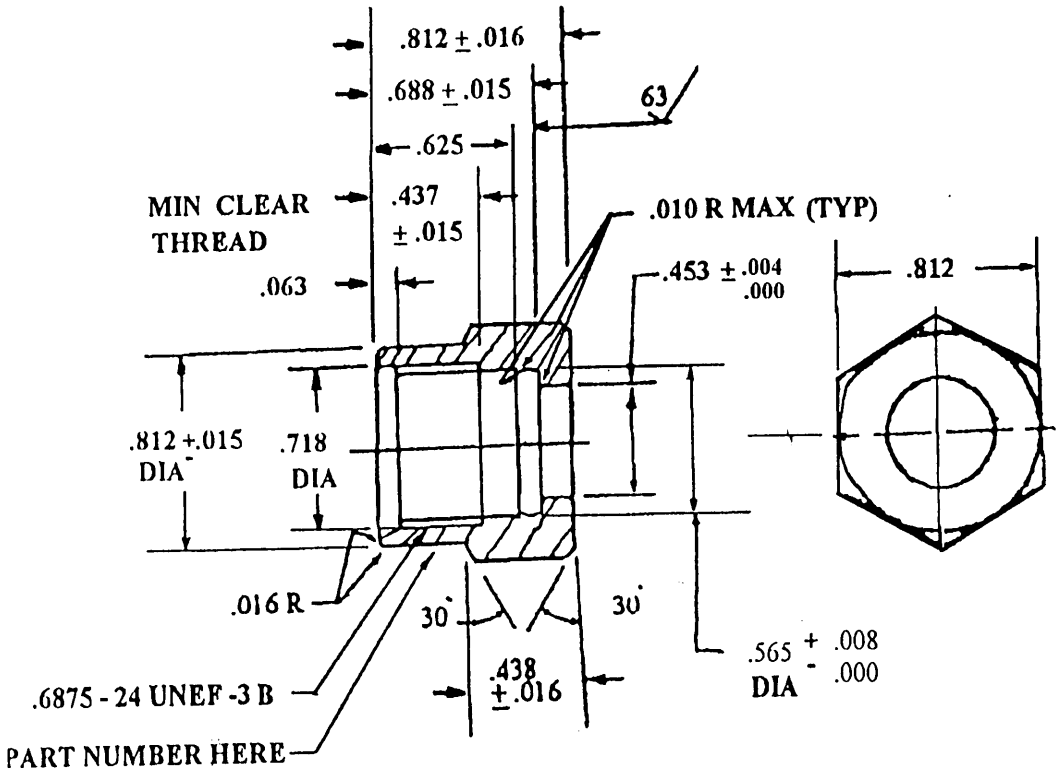
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Note: 1. 0.312 OD packing ring -6 shall be provided with each nose adapter.
2. MS 16624-4043 shall be provided with each nose adapter.
3. All dimensions are in inches.
4. Unless otherwise specified on the drawing, tolerances: decimal ± .005 and angle ± 1°.

FIGURE 2. Nose adapter.

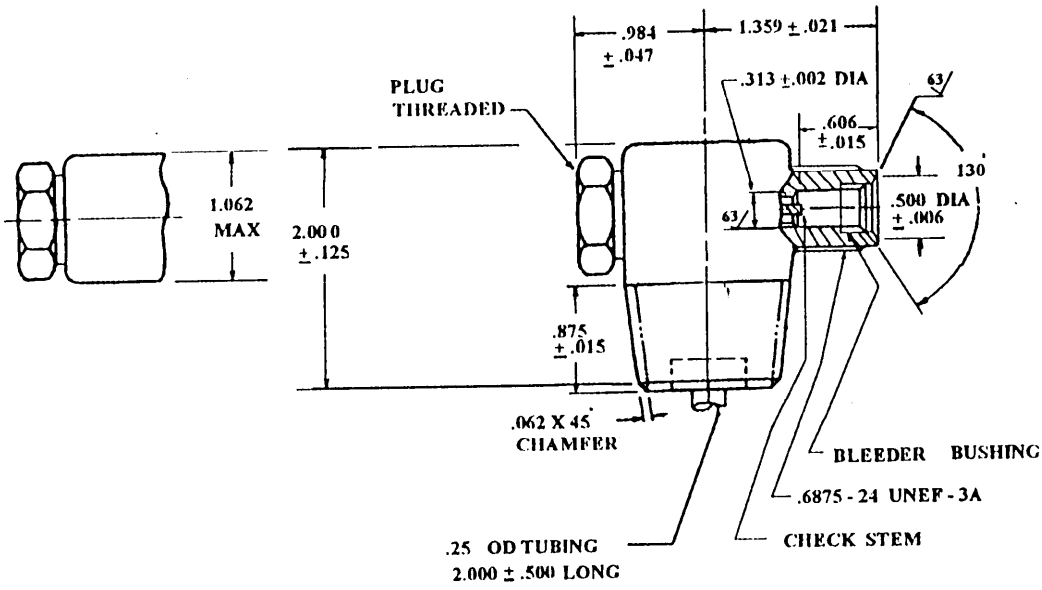
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Note: 1. All dimensions are in inches.
2. Unless otherwise specified on the drawing, tolerances: decimal ± .005 and angle ± 1°.

FIGURE 3. Coupling nut.

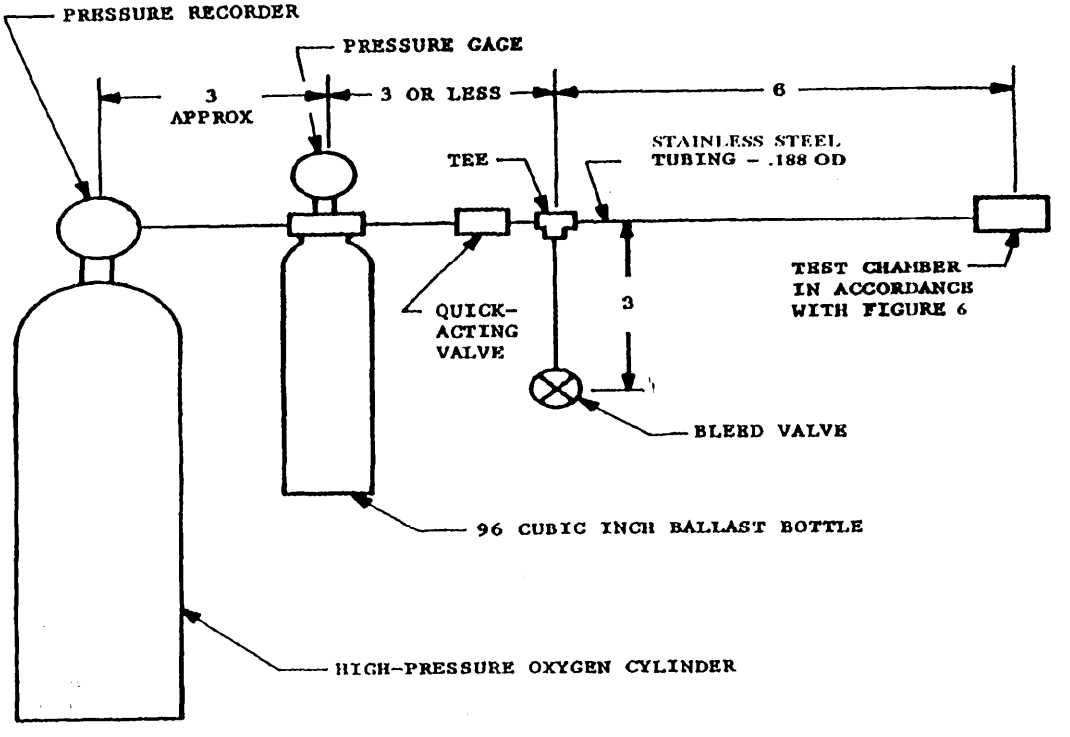
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Note: 1. All dimensions are in inches.
2. Unless otherwise specified on the drawing tolerances: decimal ± .005 and angle ± 1°.

FIGURE 4. Oxygen cylinder valve.

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Note: 1. All dimensions are in inches.
2. Unless otherwise specified on the drawing, tolerances: decimal ± .005 and angle ± 1°.

FIGURE 5. Oxygen bomb test equipment.

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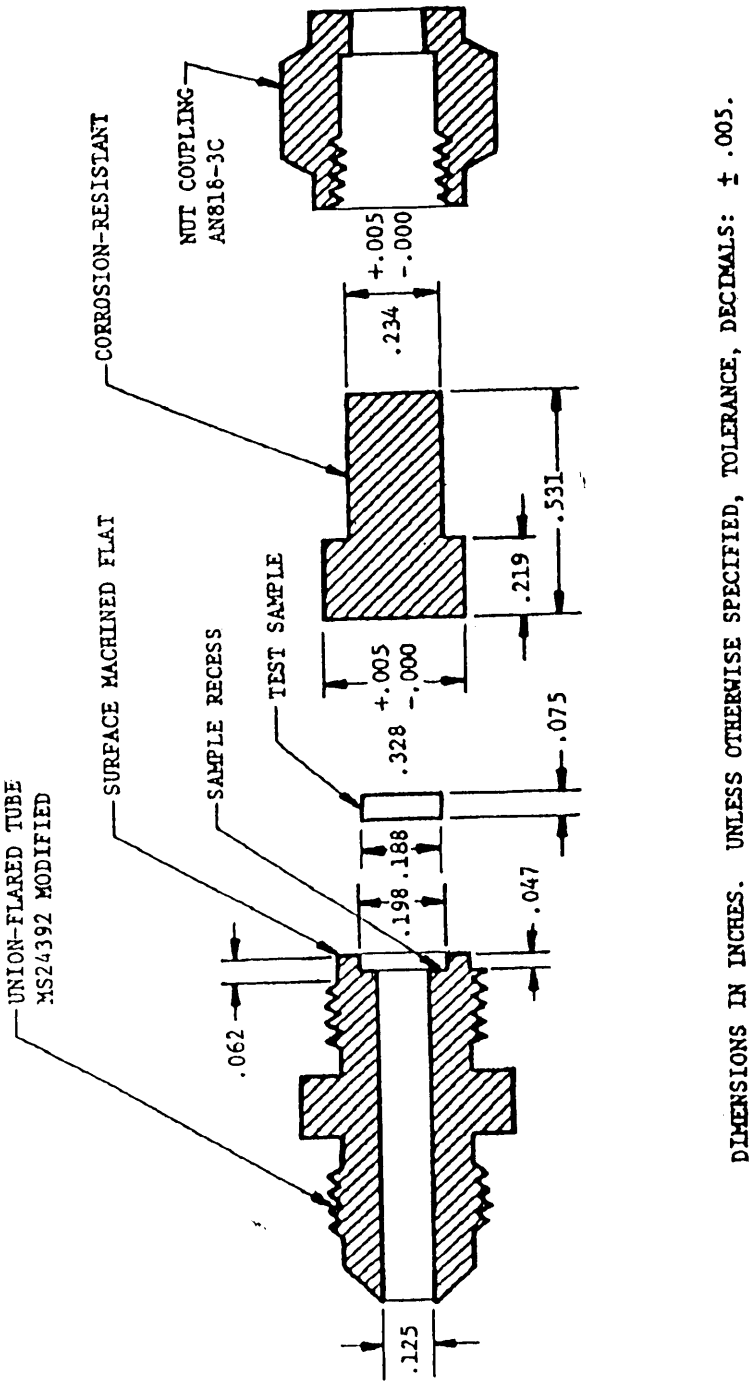


FIGURE 6. Test chamber,

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-8622C	**2. DOCUMENT DATE (YYMMDD)** 1997 January 27

3. DOCUMENT TITLE VALVE, HIGH-PRESSURE, OXYGEN CYLINDER, AUTOMATIC OPENING

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7.DATE SUBMITTED** (YYMMDD)

8. PREPARING ACTIVITY

a. NAME COMMANDER NAVAL AIR WARFARE CENTER, AIRCRAFT DIV	b. TELEPHONE *Include Area Code)* (1) Commercial (2) AUTOVON
c. ADDRESS *(Include Zip Code)* CODE SR3 HIGHWAY 547 LAKEHURST, NJ 08733-5100	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE QUALITY AND STANDARDIZATION OFFICE 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22401-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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